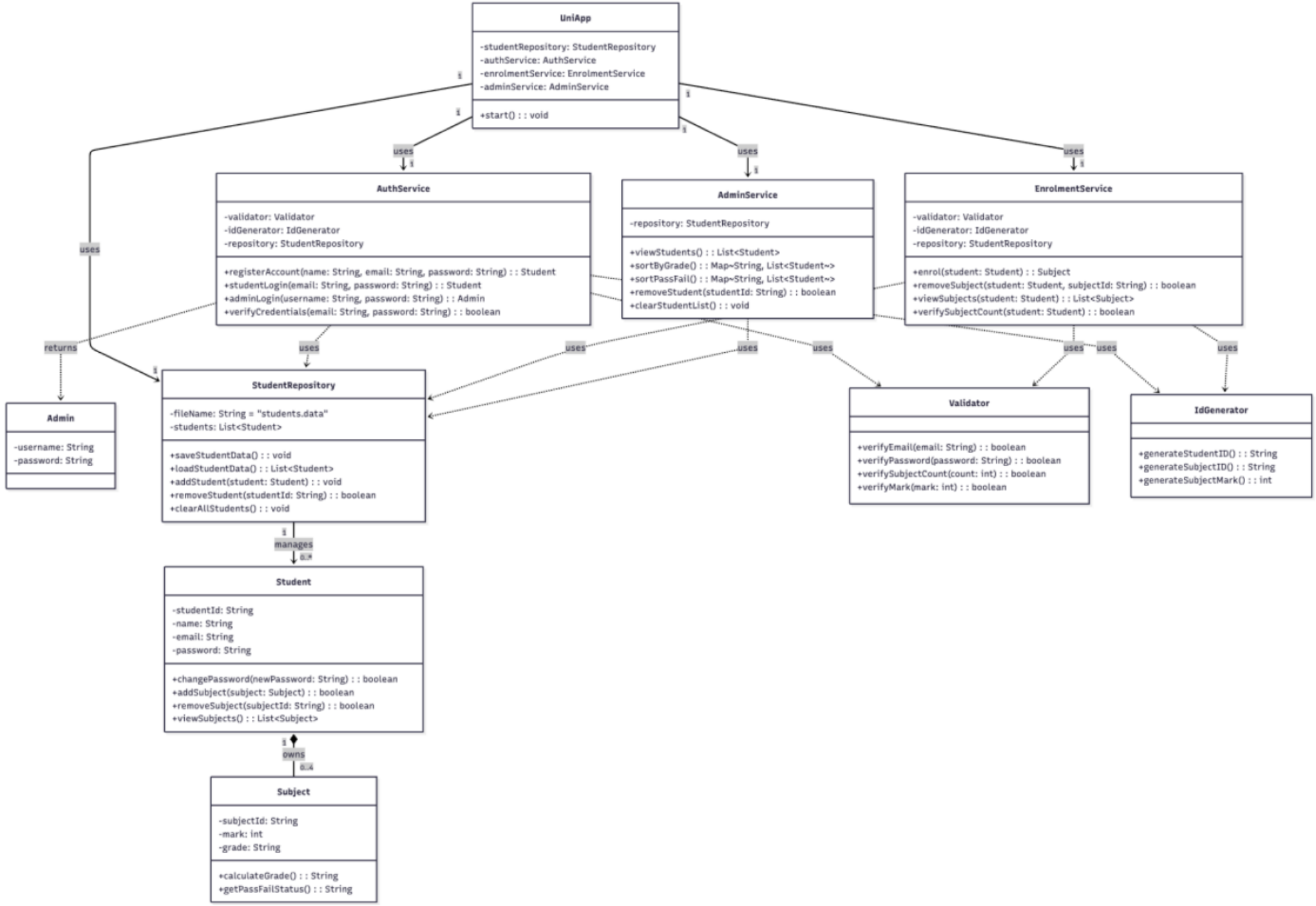

UniApp
-studentRepository: StudentRepository
-authService: AuthService
-enrolmentService: EnrolmentService
-adminService: AdminService
+start() : : void
uses
AuthService
-validator: Validator
-idGenerator: IdGenerator
-repository: StudentRepository
+registerAccount(name: String, email: String, password: String) : : Student
+studentLogin(email: String, password: String) : : Student
+adminLogin(username: String, password: String) : : Admin
+verifyCredentials(email: String, password: String) : : boolean
AdminService
-repository: StudentRepository
+viewStudents() : : List<Student>
+sortByGrade() : : Map<String, List<Student>>
+sortPassFail() : : Map<String, List<Student>>
+removeStudent(studentId: String) : : boolean
+clearStudentList() : : void
EnrolmentService
-validator: Validator
-idGenerator: IdGenerator
-repository: StudentRepository
+enrol(student: Student) : : Subject
+removeSubject(student: Student, subjectId: String) : : boolean
+viewSubjects(student: Student) : : List<Subject>
+verifySubjectCount(student: Student) : : boolean
returns
Admin
-username: String
-password: String
StudentRepository
-fileName: String = "students.data"
-students: List<Student>
+saveStudentData() : : void
+loadStudentData() : : List<Student>
+addStudent(student: Student) : : void
+removeStudent(studentId: String) : : boolean
+clearAllStudents() : : void
Validator
+verifyEmail(email: String) : : boolean
+verifyPassword(password: String) : : boolean
+verifySubjectCount(count: int) : : boolean
+verifyMark(mark: int) : : boolean
IdGenerator
+generateStudentID() : : String
+generateSubjectID() : : String
+generateSubjectMark() : : int
manages
0..*
Student
-studentId: String
-name: String
-email: String
-password: String
+changePassword(newPassword: String) : : boolean
+addSubject(subject: Subject) : : boolean
+removeSubject(subjectId: String) : : boolean
+viewSubjects() : : List<Subject>
owns
0..4
Subject
-subjectId: String
-mark: int
-grade: String
+calculateGrade() : : String
+getPassFailStatus() : : String

Main Class / GUI

Varun

University System
- Main()

GUI APP

CLI / Controller Classes

Controller Classes could inhereit from Generic Controller Class

Siddhart

University Controller
- Admin
- Student
Exit

Student Controller
- Login
- Register
- Exit

Subject Controller
- Change
- Enroll
- Remove
- View
- Exit

Admin Controller
- Clear database
- Group students
- Partition students
- Remove students
- Show
- exit

No admin class required
-> No login process, just press admin

System Classes

Student Repository

Student

Subject

Mathis

Database (.data file)

Validator

ID Generator

Shiva